

Instrumental Variable - 25 Oct 2024



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Learning Objects

1. **Why endogenous Matter?**
2. **Instrumental Variable approaches**
3. **Key conditions for a valid instrument**
4. **Identification tests of conditions**
5. **Regression results interpretations**
6. **IV in different fields**

1. Why endogenous Matter?

1. Endogenous

Violations of the condition $E[e|x]=0$

1. Omitted variable bias: see the below example
2. Measurement errors
3. Reverse causality
4. Selection bias

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

$$x_2 = \gamma_0 + \gamma_1 x_1 + \epsilon$$

$$y = \theta_0 + \theta_1 x_1 + e$$

$$\theta_1 = \beta_1 ?$$

1. Endogenous: Omitted variable bias

$$y = \theta_0 + \theta_1 x_1 + e \quad \Rightarrow$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

$$= \beta_0 + \beta_1 x_1 + e$$

$$E(y|x_1) = \beta_0 + \beta_1 x_1 + E(e|x_1)$$

$$E(e|x_1) = E(\beta_2 x_2 + \varepsilon|x_1) = \beta_2 E(x_2|x_1) + E(\varepsilon|x_1)$$

$$E(x_2|x_1) = E(\gamma_0 + \gamma_1 x_1) = \gamma_0 + \gamma_1 x_1$$

$$\Rightarrow E(e|x_1) = \beta_2(\gamma_0 + \gamma_1 x_1)$$

$$E(y|x_1) = \beta_0 + \beta_1 x_1 + \beta_2 \gamma_0 + \beta_2 \gamma_1 x_1$$

$$= \beta_0 + \beta_2 \gamma_0 + (\beta_1 + \beta_2 \gamma_1) x_1$$

$$\theta_0 = \beta_0 + \beta_2 \gamma_0$$

$$\theta_1 = \beta_1 + \beta_2 \gamma_1$$

$$\theta_1 - \beta_1 = \beta_2 \gamma_1$$

If $\beta_2 \gamma_1 > 0$: θ_1 **over estimates** β_1

If $\beta_2 \gamma_1 < 0$: θ_1 **under estimates** β_1

1. Endogenous

Solution:

If you only care about forecasting, not causal interpretation => **No actions** are needed.

Casual: Looking for exogenous variation in the exogenous variable of interest (X)

1. Add x_2 in the regression
2. **Instrumental variables (IVs)**
3. Fixes effect models (not enough)
4. Difference-in-difference estimates (exogenous shock)
5. Research discontinuity design

2. Instrumental Variable (IV) approaches

2. Instrumental Variable approaches

An instrument “**Z**” is :

$$y = \beta_0 + \beta_1 x_1 + e$$

1. Exogeneity: $E(e|Z) = 0$
2. Relevance: $E(x_1|Z) = \pi_0 + \pi_1 Z$, where $\pi_1 \neq 0$ (the stronger the better)

=> Using Z to identify the **variation of x** that is **unrelated to e**.

$$E(y|Z) = E[\beta_0 + \beta_1 x_1 + e | Z] = \beta_0 + \beta_1 E[x_1|Z] \Rightarrow \text{Correct } \beta_1$$

Method: Two-stage Least Square (2SLS)

Stage 1: Fitted value of x on Z: $\widehat{x}_1 = E(x_1|Z) = \pi_0 + \pi_1 Z$

Stage 2: Y on $\widehat{x}_1$ rather than $x_1 \Rightarrow \text{Correct } \widehat{\beta}_1$

However, standard errors (s.e.) in Stage 2 need to be adjusted (packages can do this for us).

2. Instrumental Variable approaches

Challenges of the IV approach:

Requirements: 1) Exogeneity and 2) Relevance

- Economic arguments needed for **Exogeneity**: only statistical validations are insufficient.
- In practice: **multiple IVs** (commonly two, but not too many) for each endogenous X.

2. Instrumental Variable approaches

Testing the validity of instruments:

- **Relevance** (weak identification tests): In the first stage:
 1. Highly significant π hints the relevance of IVs.
 2. One IV: 1) ***First stage F-statistics*** ≥ 10 ; 2) ***Cragg-Donald Wald F statistic***
 3. Multiple IVs: 1) ***Cragg-Donald Wald F***; 2) ***Kleibergen-Paap rk Wald (robust s.e.)***
 4. Comparing with ***Stock-Yogo critical values*** (relative bias & size)
- **Exogeneity** (Under- & Over- identification tests):
 1. **Under-identification** test: H_0 : At least one of the IVs is not exogenous.
 2. **Over-identification** test: H_0 : all IVs are exogenous.
 - needs ≥ 2 IVs for each endogenous X.

3. Real examples

3. Real examples

Acemoglu, Johnson, and Robinson (2001, AER):

- Institution quality (X) => Economy development (Y).
- logpgp95 (GDP, Y) on avexpr (property rights protection, X)
- Using 64 countries of ex-colonies (for the reason of IVs)

Endogeneity reasons: 1) reverse causality; 2) omitted variables.

Three IVs:

Idea: settler mortality=>settlements=>**early institutions**=> current institutions(X)=>economic now(Y)

1. *logem4* – Log settler mortality, based on the earliest available record.
2. *euro1900* – Fraction of the population of European descent in 1990.
3. *democ00a* – Democracy in 1900 and first year of independence (1-10, where high scores mean more democracy).

3. Real examples: Stata Outputs

AJR Ordinary least squares

$b_{OLS} = .532$; Std.

```
. reg logpgp95 avexpr
```

Stata grammar: reg Y X

Source	SS	df	MS	Number of obs	=	111
-----+-----				F(1, 109)	=	171.44
Model	88.3624725	1	88.3624725	Prob > F	=	0.0000
Residual	56.180757	109	.515419789	R-squared	=	0.6113
-----+-----				Adj R-squared	=	0.6078
Total	144.543229	110	1.31402936	Root MSE	=	.71793

logpgp95	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+-----						
avexpr	.5318713	.0406212	13.09	0.000	.4513614	.6123813
_cons	4.626089	.3005752	15.39	0.000	4.030359	5.22182

3. Real examples: Stata Outputs

One IV: *logem4* (Log settler mortality)

$b_{IV} = .868$; effect is larger but less precise.

```
. ivreg2 logpgp95 (avexpr = logem4)
```

Stata grammar: `ivreg2 Y (X=IV1 IV1 etc.)`

IV (2SLS) estimation

Estimates efficient for homoskedasticity only

Statistics consistent for homoskedasticity only

					Number of obs =	70
					F(1, 68) =	48.25
					Prob > F =	0.0000
Total (centered) SS	=	77.42744945			Centered R2 =	0.3045
Total (uncentered) SS	=	4717.418333			Uncentered R2 =	0.9886
Residual SS	=	53.85251417			Root MSE =	.8771

logpgp95	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
avexpr	.8683933	.1232135	7.05	0.000	.6268992 1.109887
_cons	2.370164	.8255731	2.87	0.004	.7520701 3.988257

3. Real examples: Stata Outputs

Multiple IVs: all three

Similar effect to before; slightly more precise. Would 2-step FEGMM help?

```
. ivreg2 logpgp95 (avexpr = logem4 euro1900 democ00a)
```

Stata grammar: `ivreg2 Y (X=IV1 IV1 etc.)`

IV (2SLS) estimation

Estimates efficient for homoskedasticity only

Statistics consistent for homoskedasticity only

		Number of obs =		59	
		F(1, 57) =		47.93	
		Prob > F =		0.0000	
Total (centered) SS	=	57.86984178	Centered R2	=	0.2722
Total (uncentered) SS	=	3850.192092	Uncentered R2	=	0.9891
Residual SS	=	42.11793954	Root MSE	=	.8449

logpgp95	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
avexpr	.8497074	.1206347	7.04	0.000	.6132678	1.086147
_cons	2.496792	.7914348	3.15	0.002	.9456088	4.047976

3. Real examples: Stata Outputs

Identification Tests under ivreg2: Under(reject), Weak(reject), Over-(fail to reject)

Underidentification test (Anderson canon. corr. LM statistic):		22.439	Need reject
Chi-sq(3) P-val =		0.0001	

Weak identification test (Cragg-Donald Wald F statistic):		11.252	Need reject
Stock-Yogo weak ID test critical values:			
	5% maximal IV relative bias	13.91	
	10% maximal IV relative bias	9.08	
	20% maximal IV relative bias	6.46	
	30% maximal IV relative bias	5.39	
	10% maximal IV size	22.30	
	15% maximal IV size	12.83	
	20% maximal IV size	9.54	
	25% maximal IV size	7.80	
Source: Stock-Yogo (2005). Reproduced by permission.			

Sargan statistic (overidentification test of all instruments):		2.015	Need fail to reject
Chi-sq(2) P-val =		0.3651	

Instrumented:		avexpr	
Excluded instruments:		logem4 euro1900 democ00a	

3. Real examples: Stata Outputs

Multiple IVs: 1)all three & 2)first stage output & 3)robust Std. Err

```
. ivreg2 logpgp95 (avexpr=logem4 euro1900 democ00a), first robust
```

...

First-stage regression of avexpr:

Statistics robust to heteroskedasticity

Number of obs = 59

Stata grammar: `ivreg2 Y (X=IV1 IV1 etc.) , first robust`

first: **also** report the 1st stage of the 2SLS.

robust: use robust s.e. to do tests.

		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
avexpr							
logem4		-.3153861	.2007578	-1.57	0.122	-.7177137	.0869415
euro1900		.0214339	.0065227	3.29	0.002	.0083622	.0345057
democ00a		.0301716	.0545556	0.55	0.582	-.0791603	.1395035
_cons		7.573125	1.048158	7.23	0.000	5.47257	9.67368

3. Real examples: Stata Outputs

Identification Tests under ivreg2, robust: Under(reject), Weak(reject)

Underidentification test			Need reject
Ho: matrix of reduced form coefficients has rank=K1-1 (underidentified)			
Ha: matrix has rank=K1 (identified)			
Kleibergen-Paap rk LM statistic	Chi-sq(3)=8.82	P-val=0.0318	
Weak identification test			Need reject
Ho: equation is weakly identified			
Cragg-Donald Wald F statistic		11.25	
Kleibergen-Paap Wald rk F statistic	Statistics using robust s.e. :17.68		
Stock-Yogo weak ID test critical values for K1=1 and L1=3:			
	5% maximal IV relative bias	13.91	
	10% maximal IV relative bias	9.08	
	20% maximal IV relative bias	6.46	
	30% maximal IV relative bias	5.39	
	10% maximal IV size	22.30	
	15% maximal IV size	12.83	
	20% maximal IV size	9.54	
	25% maximal IV size	7.80	
Source: Stock-Yogo (2005). Reproduced by permission.			
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.			

4. Useful resources

4. Useful resources

Collections of IVs:

[Sangmin Simon Oh](#)

- Assistant Professor of Finance, Columbia Business School
- [A website](#): a list of plausibly exogenous variations found in the empirical economics literature

[Sangmin Simon Oh](#)

Sangmin Simon Oh

Assistant Professor of Finance, Columbia Business School

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References

Conference Deadlines

: Collection of deadlines for major conferences in finance

Code and Data

Plausibly Exogenous Galore

: Compilation of plausibly exogenous variation used in empirical economics work (updated daily through this [listserv](#))

Variables Construction in Finance

: Compilation of norms on constructing often-used variables in finance research

High-Frequency Data

: Collection of publicly available high-frequency data

Identifiers for Major Companies

: Collection of identifiers in common datasets for a sample of large companies



Plausibly Exogenous Galore

This is a list of plausibly exogenous variations found in the empirical economics literature.
You can also sign up for a daily update [here](#).

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Default View



Citation	Title	LHS	RHS	Source of Exogenous Variation
Dyck, Morse, and Zingales (2020)	How Pervasive is Corporate Fraud?	Undetected share of corporate fraud	-	Arthur Andersen's demise that produced increased sc clients and increased the detection likelihood of pre-ex
Bernheim, Meer, and Novarro (2016)	Do consumers exploit commitment opportunities? Evidence from natural experiments involving liquor consumption	Alcohol consumption	"Availability strategy" of regulating consumption	State-level policy changes that altered the hours durin can purchase liquor on Sundays (so-called "blue laws"
Choi, Guzman, and Small (2024)	Third Places and Neighborhood Entrepreneurship: Evidence from Starbucks Cafés	Entrepreneurship	"Third places" (e.g. neighborhood cafes)	Introduction of Starbucks cafés into U.S. neighborhood shops
Choi, Guzman, and Small (2024)	Third Places and Neighborhood Entrepreneurship: Evidence from Starbucks Cafés	Local entrepreneurship	"Third places" such as neighborhood cafes	Introduction of Starbucks cafés into U.S. neighborhood shops
Bhuller et al. (2018)	Incarceration Spillovers in Criminal and Family Networks	Probability that one's criminal network members or brothers is charged	A defendant is sent to prison	Quasi-random assignment of criminal cases to judges

Summary

1. Endogenous

- => biased coefficients estimations
- Can be used to predict but $\neq$ casual interpretations

2. Valid instrument:

- 1) Exogenous;
- 2) Relevance (strong)

3. Identification tests:

- 1) under- & Over-, and
- 2) weak

4. Regression results:

- 1) reject & fail to reject, and
- 2) reject
- Otherwise => Change IV